Nasir Sims

nasirsims.dev

Education

Washington University in St. Louis — St. Louis, MO

Bachelor of Science in Computer Science, Minor in Film & Media Studies — *May 2027*

Relevant Courses: Operating Systems, Introduction to Artificial Intelligence, Object-Oriented Programming, Parallel & Concurrent Programming, Programming Systems & Languages, Digital Logic & Computer Design, Data Structures & Algorithms

Projects

Bidirectional Camera System | *Raspberry Pi, Bash, Soldering* — Nov. 2024 - Apr. 2025

- Led the redesign of the rocket's onboard camera system, developing a custom Raspberry Pi Zero 2W-based dual-camera architecture to capture forward and aft flight footage for post-flight analysis and airbrake verification
- Streamlined post-flight data recovery by implementing local video encoding and optimized transfer workflows, while collaborating with avionics and electrical subteams to validate power reliability, and successful data capture during launch

AV Intersection Detection Classification Model | *CARLA, Python, TensorFlow* — Dec. 2024 - Apr. 2025

- Constructed a balanced dataset of 700+ images on lane markings and intersections from CARLA environment simulations to model synthetic data, with labels for appropriate intersection directions
- Trained a ML classification learning model to analyze left, right, and straight intersection lanes and execute autonomous maneuvering based on intersection classifications

Telemetry Base Station | *RF Signal Processing, Python, C, LoRa* — Oct. 2023 - Apr. 2024

- Engineered a LoRa telemetry system with Adafruit Feather M0 microcontrollers and custom 915MHz RF channels, enabling real-time bidirectional communication and transmission of over 20+ critical flight variables
- Developed a serial interface for real-time execution of drone payload deployment during critical launch phases

Experience

Chief Safety Officer, WURocketry — Sep. 2023 – Present

Avionics Lead (Jan. – Apr. 2025), Avionics Member (Sep. 2023 – Dec. 2024) — *St. Louis, MO*

- Collaborate with a team of over 80 people to develop, build, and launch a rocket for the NASA USLI Division
- Serve as the primary liaison between WURocketry and NASA safety officials, ensuring compliance with NASA Student Launch safety requirements, policies, and strict documentation standards
- Built and maintain an avionics software library to standardize sensor methods using abstract class architecture

Drone Pilot Intern — May 2025 – Aug. 2025

Washington University in St. Louis Facilities Planning & Management — *St. Louis, MO*

- Piloted DJI Matrice 300 RTK and Mavic 3, processing aerial imagery in Esri Drone2Map to generate orthomosaics, DSMs, and classified point clouds for downstream geospatial analysis
- Executed LiDAR classification and raster analysis in ArcGIS Pro, applying SLAM-supported workflows to refine ground, building, and vegetation classes for accurate change detection, including seasonal foliage variation and post-tornado damage
- Led the end-to-end internal space audit program, combining Insta360 cameras, ArcGIS Pro, and FME to create the geospatial foundation for a campus-scale digital twin and support business data driven decisions

Research Assistant — Aug. 2024 – Apr. 2025

Independent Study, Washington University in St. Louis — *St. Louis, MO*

- Worked with a multidisciplinary team of researchers, conducting rigorous stress-testing of Autonomous Vehicles in simulated environments to enhance safety, address vulnerabilities, and improved reliability for real-world deployment
- Interfaced with a ROS 2-based autonomous navigation framework integrating LaneNet, sensor fusion, and an actor-critic reinforcement learning agent with a neural network employing PPO for lane-following and obstacle avoidance in CARLA simulations and a 1:8-scale urban model, achieving stable performance across diverse traffic and environmental conditions
- Customized LaneNet using TensorFlow to enhance lane detection at intersections, significantly improving accuracy

InfoSec GRC Analyst Intern — Apr. 2024 – Present

Washington University in St. Louis — *St. Louis, MO*

- Deliver timely updates to information security policies to meet compliance standards and audit schedules, ensuring consistent security practices across all university departments
- Lead risk assessments for WashU's 1000+ vendor network, utilizing OneTrust and Black Kite security platforms to streamline compliance, reduce cybersecurity incidents, and adhere to evolving AI-driven industry standards
- Manage and update a glossary of 200+ IT terms to maintain accurate documentation for organizational use

Skills

Languages: C++, C, Java, Python, Bash, SML, AVR/RISC-V Assembly, Ruby, Racket, Intermediate Mandarin Chinese
Software: ArcGIS Pro, FME, Esri Drone2Map, GeoSLAM, Pix4D, Office 365, MySQL, Tableau
Libraries: CARLA, PyTorch, TensorFlow, Gymnasium, scikit-learn, BSXlite, RadioHead, Pandas, Matplotlib, NumPy, Eigen